MANI SHARAN SANDELA

Southfield, MI | +1 (248-730-5508) | manisharan0902@gmail.com | Microsoft Certified: Azure Data Engineer Associate (DP-203) | AWS Certified Data Engineer - Associate | Databricks Data Engineer Associate

PROFESSIONAL SUMMARY

Results-driven Data Engineer with 5 + years of experience designing and delivering cloud-native data platforms on Azure and AWS. Specialized in building scalable ETL/ELT pipelines, Lakehouse architectures, and real-time streaming solutions processing multi-terabyte datasets. Proven expertise in Python, PySpark, SQL, Databricks, dbt, Azure Data Factory, AWS Glue, Apache Kafka, Airflow, and Terraform. Certified Azure Data Engineer, AWS Data Engineer, and Databricks Data Engineer. Strong background in financial services and healthcare domains with demonstrated impact on data quality, pipeline performance, and operational cost reduction.

TECHNICAL SKILLS

Languages: Python, SQL
Big Data & Processing: Apache Spark, PySpark, Databricks, Distributed Computing
Data Engineering: ETL, ELT, dbt, Data Modeling, Dimensional Modeling, Star Schema, CDC (Change Data Capture)
Lakehouse & Architecture: Delta Lake, Medallion Architecture, Lakehouse Architecture, Unity Catalog, Data Vault
Streaming & Messaging: Apache Kafka, AWS Kinesis, Azure Event Hub, Event-Driven Architecture, Real-Time Pipelines
Cloud – Azure: Azure Data Factory (ADF), Azure Databricks, Azure Data Lake Storage Gen2, Azure Key Vault, Azure OpenAI
Cloud – AWS: AWS Glue, Amazon S3, Amazon Redshift, AWS Lambda, Amazon Athena, AWS Kinesis
Orchestration & DevOps: Apache Airflow, Terraform, Azure DevOps, GitHub Actions, Docker, Kubernetes, CI/CD
Data Quality & Governance: Great Expectations, Data Lineage, Metadata Management, IAM, Encryption, HIPAA, HL7 **BI & Analytics:** Power BI, Tableau, Analytics Engineering

PROFESSIONAL EXPERIENCE

Data Engineer

Capital One Financial | McLean, VA (Remote) Sep 2025 – Present

- Architected 45+ ADF, Databricks, dbt, and PySpark ETL/ELT pipelines processing 8TB+ of daily financial data across 10+ business units, reducing reporting latency by 60% and expanding analytics access for 200+ stakeholders.
- Designed and implemented a 50TB+ Azure Data Lake Storage Gen2 Lakehouse platform using Delta Lake, Medallion Architecture, and Unity Catalog, reducing data retrieval times by 45% and strengthening data governance and lineage.
- Optimized large-scale PySpark workloads processing billions of records monthly, cutting execution times by 35%, achieving 99.9% SLA compliance, and lowering cloud compute costs significantly.
- Built real-time data ingestion frameworks using Azure Event Hub, Apache Kafka, and Delta Lake to support enterprise analytics and operational reporting.
- Developed AI-enabled data products integrating Azure OpenAI for automated data classification, enrichment, and metadata generation, reducing manual processing effort by 70%.
- Implemented enterprise-wide data quality and observability framework using Great Expectations across 500+ datasets, reducing production data defects by 60%.
- Automated cloud infrastructure provisioning via Terraform and CI/CD pipelines (Azure DevOps, GitHub Actions, Docker), reducing deployment effort by 40%.

Data Engineer

Tata Consultancy Services | India (Remote) Mar 2021 – July 2023

- Migrated 30+ enterprise ETL/ELT workflows from on-premises legacy systems to AWS Glue, Amazon S3, Amazon Redshift, Apache Airflow, and dbt, reducing operational costs by 30% and improving scalability.
- Built and maintained 50+ PySpark, SQL, and dbt transformation pipelines processing 500M+ records monthly, enabling enterprise-wide reporting and analytics.
- Architected real-time streaming platforms using Apache Kafka, AWS Kinesis, and event-driven design across 20+ data streams, reducing data delivery latency by 40%.
- Optimized Amazon Redshift query performance through advanced tuning, partitioning strategies, and dimensional modeling, improving analytical query performance by 25%.
- Developed and maintained 40+ Apache Airflow DAGs orchestrating critical production workflows with reliable scheduling, monitoring, and dependency management.
- Implemented CDC-based ingestion frameworks and metadata-driven pipeline architectures, improving engineering productivity by 20%.

- Enforced data governance, security, and compliance through IAM policies, encryption standards, data lineage tracking, and access-control frameworks.

Associate Data Engineer

Max Healthcare | India

May 2019 – Feb 2021

- Developed Python and SQL ETL pipelines to integrate healthcare data from multiple source systems, ensuring full HL7 and HIPAA compliance.
- Built Spark-based transformation workflows processing millions of healthcare records, improving reporting performance by 20%.
- Designed dimensional data models and curated data warehouse layers supporting analytics and operational reporting workloads.
- Implemented automated monitoring, validation, and reconciliation frameworks to improve data quality, reliability, and reporting accuracy.

EDUCATION

Master of Science, Information Technology

Lawrence Technological University — Southfield, MI